Contributions and evidence boundaries

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Contributions and evidence boundaries

We make ten contributions, each paired with a theorem, figure, table, or generated token and with an explicit boundary on what the evidence establishes.

Contributions and evidence... (part 2)

They fall into four groups: the first two build the core and its recovery contract; the next two report and statistically qualify the contaminated-consensus result; the fifth and sixth make the robustness axes explicit and add the objective-backed server rule; and the remaining four are scoped extensions —

Contributions and evidence... (part 3)

parameter recovery, the tempered aggregation family, an aggregation-API transfer, and a disjoint-observation communication test.

1. A discrete-categorical FedGVI core

A typed, deterministic, pure-NumPy/SciPy reimplementation of the FedGVI (Mildner et al. 2025) generalized-Bayes primitives — divergences, bounded robust losses, the generalized posterior, the cavity/factor algebra, and robust aggregation

—

1. A discrete-categorical... (part 2)

in the discrete-categorical setting that active inference (Da Costa et al. 2020) uses.

1. A discrete-categorical... (part 3)

The objective (eq. 2) and its closed-form tempered-softmax solution are stated in Definition 1 and tested by the recovery-limit probes. The whole core is zero-mock and reproducible (Peng 2011).

2. A recovery-tested connection between categorical pooling and robust Bayes

2. A recovery-tested... (part 2)

A recovery certificate shows that the client KL/negative-log-likelihood loss limits recover Bayes and the server's zero-robustness branch recovers the project log-linear pool.

2. A recovery-tested... (part 3)

Under the explicit shared-support, posterior-log-potential, and fixed-weight assumptions in sec. 8.8, that pool specializes Friston et al.'s Eq. 7 message-combination term (Friston et al. 2024), not the complete source protocol.

2. A recovery-tested... (part 4)

Three recovery limits are pinned to bit-level residuals. The bounded losses recover NLL/Bayes (Corollary 6 + Proposition 4; ≤ 0 and ≤ 0 maximum residual, eq. 11).

2. A recovery-tested... (part 5)

The Rényi divergence recovers KL (Lemma 3; ≤ 0 residual, eq. 6). The server-side reweighting pool recovers the naive pool in its trusting limit (Theorem 5; ≤ 0 residual, eq. 10).

2. A recovery-tested... (part 6)

Robustness is thereby a tested recovery-limit extension, not a replacement.

3. End-to-end evaluation of robust federated active inference

Three worked categorical source-mechanism analogues cover communicating colonies reaching lower free energy (sec. 10.2), Dirichlet language acquisition (sec. 10.3),

3. End-to-end evaluation of... (part 2)

and a configured Bayesian-model- reduction pruning sign control (Friston and Penny 2011) (sec. 10.4).

3. End-to-end evaluation of... (part 3)

A contaminated-sentinel robustness sweep (sec. 10.5) shows the naive pool degrading while at least one server-side robust member clears the configured threshold at the most severe swept rate.

4. A statistically qualified server-side contrast

The “robust beats naive” conclusion for the declared contamination rate is produced only by a matched-pairs Wilcoxon signed-rank test (Wilcoxon 1945).

4. A statistically qualified... (part 2)

Its multiplicity correction uses Benjamini–Hochberg FDR (Benjamini and Hochberg 1995) across the divergence family.

We report the contrast with bootstrap confidence intervals (Efron and Tibshirani 1993) and observed-effect design-power planning.

4. A statistically qualified... (part 3)

Across 960 paired trials, the headline display method (RKL; tied set: RKL, AR, beta, rcce) reaches accuracy 0.9829 against the naive pool's 0.9021 at the verdict rate.

4. A statistically qualified... (part 4)

At that rate, $q = 1.11 \times 10^{-158}$, and the rank-biserial-derived d-equivalent is saturated (r=+1). The predeclared selection rule is largest positive rank-biserial effect_size; stable method order tie-break; the method with the largest paired mean difference is AR. Every headline number is a generated token.

5. An explicit accounting of the robustness axes

The map in sec. 6.3 separates the client-side per-agent update, which inherits FedGVI's bounded-influence result under its stated source assumptions, from both server rules.

5. An explicit accounting of... (part 2)

The sharp server-side reweighting heuristic has a recovery limit as its positive formal property and a scoped no-go result for its declared separable objective class.

5. An explicit accounting of… (part 3)

The conservative variational server rule is objective-backed and has a redescending effective-weight update, but is not the accuracy-maximizer. Downstream users are told exactly which result is theoretically backed and which is a labeled heuristic.

6. An objective-backed server aggregator with redescending weights

We derive an aggregation free energy eq. 12 whose exact block updates define the `variational_aggregate` rule (sec. 8.8.2, sec. 15).

6. An objective-backed... (part 2)

Each exact block update monotonically decreases a stated objective, and a converged fixed point is coordinatewise stationary. The implementation keeps the lowest observed objective among converged configured starts,

6. An objective-backed... (part 3)

or reports the best unfinished trace as non-converged.

6. An objective-backed… (part 4)

The rule recovers the standard log-linear pool in the trusting limit (eq. 10) and — unlike the sharp heuristic — carries a proven raw effective-weight bound (fig. 19, fig. 20).

6. An objective-backed... (part 5)

The honest cost is conservatism: it is a maximum-entropy-biased consensus and trades peak point-accuracy for that control, so it complements rather than replaces the sharp heuristic of contribution 5.

7. Executed finite-grid acuity-recovery experiment

At each value in the 0.60, 0.70, 0.80, 0.90 acuity grid, the study generates 200 synthetic observations in each of 960 trials. It selects acuity by marginal-likelihood grid search over the declared finite grid.

7. Executed finite-grid... (part 2)

The observed mean absolute error is 0.0232 with $R^2 = 0.9999$ (fig. 34). Acuity-by-colony-size behavior belongs to the separate sensitivity study; it is not a parameter-recovery result.

8. The tempered aggregation family

A one-parameter $\lambda > 0$ generalization of the variational aggregate (sec. 15.5) is the objective

8. The tempered… (part 2)

$$F_\lambda(q, a) = \sum_n a_n \,\mathrm{CE}(q, q_n) - \lambda H(q) + \frac{1}{c}\,\mathrm{KL}_{\mathrm{gen}}(a \| w). \tag{1}$$

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8. The tempered… (part 3)

In eq. 1, at $\lambda = 1.0$, the temperature is unity and the objective reduces to the standard variational aggregate bit-for-bit. Lower λ sharpens the variational q-block toward a maximizing state; it does not algebraically recover `robust_aggregate`.

8. The tempered… (part 4)

The raw effective-weight update and its bound are preserved for all λ. The full derivation is in sec. 15.5.

9. An aggregation-API transfer demonstration

The same `robust_aggregate` API that governs the POMDP studies is exercised unchanged with one deterministic MLP trained with the density-power β-loss (16 hidden units, $\beta = 0.5$;

9. An aggregation-API... (part 2)

generalized variational inference with a point-mass variational family) as the per-client model.

9. An aggregation-API... (part 3)

This supports portability of the server API to one additional model class (sec. 10.6) when the optional `torch` extra is installed (sec. 8.26). Without it, the MLP run is skipped and its tokens render accordingly.

10. Communication contrast under disjoint observations

A multi-agent extension (sec. 17.7.1) in

10. Communication contrast... (part 2)

which 3 agents each observe a 2-slot disjoint window shows that belief sharing materially improves over isolated-agent accuracy in the declared configuration.

10. Communication contrast... (part 3)

Isolated agents clear the 0.167 chance baseline but stay far below the communicating consensus, which itself remains well short of full accuracy.

10. Communication contrast... (part 4)

Across 128 seeds, isolated accuracy is 0.326 versus communicating 0.493, a reproducible margin under the declared matched-seed comparison (Wilcoxon $p = 9.35 \times 10^{-23}$). This is evidence for the configured disjoint-observation protocol, not a universal communication theorem.

10. Communication contrast... (part 5)

The remainder of the paper proceeds as follows. sec. 8 develops the FedGVI core and the recovery limits; sec. 9 states the numbered recovery theorems and the expected-free-energy identity; sec. 8.22 fixes the configuration;

10. Communication contrast... (part 6)

sec. 10 reports the 9 studies, beginning with the recovery checks (sec. 10.1).

10. Communication contrast... (part 7)

sec. 11 and sec. 12 synthesize; and sec. 13 and sec. 11.15 document determinism, scope, limitations, and the standing of each robustness axis.